Steps to follow for encrypting, encoding and signing the search and download API request

Whenever a new FI is registered, FI user should login and upload public key in CKYC application from User Master -> Upload Public Key page [Extract public key information from FIs private key and upload the same in CKYC upload public key page] This public key uploaded by FI will be validated and verified in step 10.

The above public key uploaded by user should be in .cer format and IP address from which user is going to hit API request needs to be selected. This public key will be stored in CKYC application in the below format.

Ex: FICODE_pub.cer

User should download CKYC public key from CKYC portal and should use the downloaded key during request encryption.

<u>The below are the steps to sign the API request:</u>

Sample: - Actual request received from consumer.

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>

<PID_DATA>

<ID_NO>D4567890</ID_NO>

<ID_TYPE>A</ID_TYPE>

</PID_DATA>

1) Generate the current date timestamp and convert it to required format using java.util library.

DateTimestamp:20-05-2021 17:15:57

2) Add this date timestamp in DATE_TIME tag of PID_DATA tag.

Data:

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>

<PID_DATA>

<DATE_TIME>20-05-2021 17:15:57</DATE_TIME>

<ID_NO>D4567890</ID_NO>

<ID_TYPE>A</ID_TYPE>

</PID_DATA>

3) Generate a random 256-bit session key with AES and BC (Bouncy castle) algorithm using javax.crypto.KeyGenerator library

Session key in bytes:[B@663c9e7a

4) Encrypt the Data(bytes) using the session key by AES Symmetric algorithm - Used org.bouncycastle.crypto.paddings.PaddedBufferedBlockCipher Cipher library

Encrypted PID in bytes: [B@606e4010

5) Encode the encrypted PID to Base64 string using org.apache.commons.codec.binary library

Encoded PID:
h5NbEOYPgJEKhhSmRpWIF58EWJgBCL5YlJYEZF1b4sbtHmhOaW4bAeX96wVq9cSdj9G
HPtDo7rTmL+5YhbVduxS0/rmOHgA5mGjRvu+oZlEzD2QxPUHeKyzOgdjR6mTUP/igXiEe7
dAwbep9h9YKQhxCmpXkWMx7VhRB+R0VkSGe5p1YctJD4copxPScHdLHROdE2YIZysSv
n8/zJy87a5f8I/yz1UIRLrKmLancHQc=

6) If version is 1.2,

Encrypt the session key using public key provided by cersai using RSA Asymmetric algorithm - Request should be encrypted with RSA/NONE/OAEPWithSHA1AndMGF1 Padding algorithm and javax.crypto.Cipher library

If version is 1.3,

Encrypt the session key using public key provided by cersai using RSA Asymmetric algorithm - Request should be encrypted with RSA/NONE/OAEPWithSHA256AndMGF1 Padding algorithm and javax.crypto.Cipher library

V1.0 or V1.1 is no longer supported.

Encrypted SessionKey in bytes:[B@21d03963

7) Encode the encrypted SessionKey to Base64 string using org.apache.commons.codec.binary library.

Encoded Session Key:

KnPPKUTjPolQ6qEWAUE4gRiz6aomJNwzUUiMFtYDJ4Q9VUuLFRMGK6eIruC2VltSDrms
Eg4zBziZnE9GmeBbGRPqLYzn2tYus+gx1TmbisxRqUCl6jt2uyJspR4+k38MeWQYsfPOtv6
VM3N2uOOhewMaQ35qBK8kKn8EuzKP4ouNYTdXxvJgmCfrwRPrHZ621wkip910dISpAH
jJQNHwL/tDEY+YMKcWMnO1CupjKBB8uIiED0pVE4W0eULfmSPEFMuCPedwuiwoQQ8t
wj3pqCUYJ4NisAvwoCkRUHPbSdkwcaCfvSbzCY4vE4v21RB3KT6MPYeWa7rC3sDOQLz
a0w==

8) Add this encrypted and encoded "PID Data" and "Session Key" to the request XML.

<?xml version="1.0" encoding="UTF-8"?><REQ_ROOT>

<HEADER>

<FI_CODE>IN1895</FI_CODE>

<REQUEST_ID>02</REQUEST_ID>

<VERSION>1.2</VERSION>

</HEADER>

<CKYC_INQ>

<PID>h5NbEOYPgJEKhhSmRpWIF58EWJgBCL5YlJYEZF1b4sbtHmhOaW4bAeX96wVq9cSdj9GHPtDo7rTmL+5YhbVduxS0/rmOHgA5mGjRvu+oZlEzD2QxPUHeKyzOgdjR6mTUP/igXiEe7dAwbep9h9YKQhxCmpXkWMx7VhRB+R0VkSGe5p1YctJD4copxPScHdLHROdE2YlZysSvn8/zJy87a5f8I/yz1UIRLrKmLancHQc=</PID>

<SESSION_KEY>KnPPKUTjPolQ6qEWAUE4gRiz6aomJNwzUUiMFtYDJ4Q9VUuLFRMGK6eIruC2VltSDrmsEg4zBziZnE9GmeBbGRPqLYzn2tYus+gx1TmbisxRqUCl6jt2uyJspR4+k38MeWQYsfPOtv6VM3N2uOOhewMaQ35qBK8kKn8EuzKP4ouNYTdXxvJgmCfrwRPrHZ621wkip910dISpAHjJQNHwL/tDEY+YMKcWMnO1CupjKBB8uIiED0pVE4W0eULfmSPEFMuCPedwuiwoQQ8twj3pqCUYJ4NisAvwoCkRUHPbSdkwcaCfvSbzCY4vE4v21RB3KT6MPYeWa7rC3sDOQLza0w==</SESSION_KEY>

</CKYC_INQ>

</REQ_ROOT>

9) Sign the above request using FI's private key [Use class2 or class3 certificate, self-signed certificate is not accepted. This private key and the key which was used to upload public key in CKYC upload public key page should be same]

First convert the request XML which is in string to document using javax.xml.parsers.DocumentBuilderFactory and org.w3c.dom.Document libraries

Get the private key details using java.security.KeyStore.PrivateKeyEntry library

Example:

Subject: CN=DS TATA CAPITAL FINANCE SERVICES LIMITED 1, OID.2.5.4.51=11TH FLOOR TOWER A

PENINSULA BUSINESS PARK SENAPATI BAPAT MARG TOWER PAREL WEST, STREET=MUMBAI,

ST=MAHARASHTRA, OID.2.5.4.17=400013, OU=LEGAL AND COMPLIANCE, O=TATA CAPITAL FINANCE

SERVICES LIMITED, C=IN

Signature Algorithm: SHA256withRSA, OID = 1.2.840.113549.1.1.11

Key: Sun RSA public key, 2048 bits

modulus:

2555472365872103366976660303167610275331545311918299946952221985239688
8031213442606922695145537145349222788712357359923953068301886656727177
3704336402205362242246589593277084474619755027677787382712635367791669
1451022840913523151714345918483129055312964461715805176323609061104626
4419804148175071840621426596234081696948290073940327832428380924190107
0758526811004352266831357133277327995415051970641423285261182792344170
4835113156112535823592288698369198111559289920489113873115410554884094
8808937787043179252509809103950944136625895309875111355737604939779022
4137956790656755345048496556302007018094945102604530538 73

public exponent: 65537

Validity: [From: Thu Jul 18 12:24:45 IST 2019,

To: Sat Jul 17 12:24:45 IST 2022]

Issuer: CN=e-Mudhra Sub CA Class 2 for Document Signer 2014, OU=Certifying Authority, O=eMudhra

Consumer Services Limited, C=IN

SerialNumber: [01067999]

Sign the document with the key using javax.xml.crypto.dsig.XMLSignatureFactory, javax.xml.crypto.dsig.SignedInfo and javax.xml.crypto.dsig.XMLSignature libraries.

Convert the document back to the string.

Below is the sample output after signing the request XML V1.2

<?xml version="1.0" encoding="UTF-8" standalone="no"?><REQ_ROOT>

<HEADER>

<FI_CODE>IN1895</FI_CODE>

<REQUEST_ID>02</REQUEST_ID>

<VERSION>1.2</VERSION>

</HEADER>

<CKYC_INQ>

<PID>h5NbEOYPgJEKhhSmRpWIF58EWJgBCL5YUYEZF1b4sbtHmhOaW4bAeX96wVq9c
Sdj9GHPtDo7rTmL+5YhbVduxS0/rmOHgA5mGjRvu+oZlEzD2QxPUHeKyzOgdjR6mTUP/i
gXiEe7dAwbep9h9YKQhxCmpXkWMx7VhRB+R0VkSGe5p1YctJD4copxPScHdLHROdE2
YlZysSvn8/zJy87a5f8I/yz1UIRLrKmLancHQc=</PID>

<SESSION_KEY>KnPPKUTjPolQ6qEWAUE4gRiz6aomJNwzUUiMFtYDJ4Q9VUuLFRMGK6
eIruC2VltSDrmsEg4zBziZnE9GmeBbGRPqLYzn2tYus+gx1TmbisxRqUCl6jt2uyJspR4+k38
MeWQYsfPOtv6VM3N2uOOhewMaQ35qBK8kKn8EuzKP4ouNYTdXxvJgmCfrwRPrHZ621
wkip910dISpAHjJQNHwL/tDEY+YMKcWMnO1CupjKBB8uIiED0pVE4W0eULfmSPEFMuC
PedwuiwoQQ8twj3pqCUYJ4NisAvwoCkRUHPbSdkwcaCfvSbzCY4vE4v21RB3KT6MPYe
Wa7rC3sDOQLza0w==</SESSION_KEY>

</CKYC_INQ>

<Signature
xmlns="http://www.w3.org/2000/09/xmldsig#"><SignedInfo><CanonicalizationMethod

Algorithm="http://www.w3.org/TR/2001/REC-xml-c14n-20010315"/><SignatureMethod

Algorithm="http://www.w3.org/2000/09/xmldsig#rsa-sha1"/><Reference

URI=""><Transforms><Transform
Algorithm="http://www.w3.org/2000/09/xmldsig#envelopedsignature"/></Transforms>
<DigestMethod

Algorithm="http://www.w3.org/2000/09/xmldsig#sha1"/><DigestValue>xTMdnPXRctasX
p0ULxT/QzOfp

90=</DigestValue></Reference></SignedInfo><SignatureValue>jqGYE7V5QcwWFzuLi3
E5XZaB/LgYK5Km

RU0Hy7T0sszo4UcS6NeetH36Iez/PnW9N2tYyKyfvBxY4FuIywOzwdeFjXTizaItYsIfBKBPev
Bq7u8lc44DkrwT

G7gJ06UqkkXzeoxje54bdlPSbl9adXeH5AIM2geh3WB/snrus7ZyZCSAng9H6LrY7F3AJJCe
55S/HpBxmgCran

93vYu2FpBJ4iK3rkxuJwwMWcwD7CG2tWxMr66UyOzTSirhs8995ceByzQHp2LiBnghFIm
ytyR4PryIcG1lDJC

/tYgAe3ZdVZZ4TvX6tjKRWOZiGBwdHQXQcEFCotDxtVbVYqyKVA==</SignatureValue></
Signature></REQ_ROOT>

Below is the sample output after signing the request XML V1.3

<?xml version="1.0" encoding="UTF-8" standalone="no"?><REQ_ROOT>

<HEADER>

<FI_CODE>IN1895</FI_CODE>

<REQUEST_ID>02</REQUEST_ID>

<VERSION>1.2</VERSION>

</HEADER>

<CKYC_INQ>

<PID>h5NbEOYPgJEKhhSmRpWIF58EWJgBCL5YlJYEZF1b4sbtHmhOaW4bAeX96wVq9cSdj9GHPtDo7rTmL+5YhbVduxS0/rmOHgA5mGjRvu+oZlEzD2QxPUHeKyzOgdjR6mTUP/igXiEe7dAwbep9h9YKQhxCmpXkWMx7VhRB+R0VkSGe5p1YctJD4copxPScHdLHROdE2YlZysSvn8/zJy87a5f8I/yz1UIRLrKmLancHQc=</PID>

<SESSION_KEY>KnPPKUTjPolQ6qEWAUE4gRiz6aomJNwzUUiMFtYDJ4Q9VUuLFRMGK6eIruC2VltSDrmsEg4zBziZnE9GmeBbGRPqLYzn2tYus+gx1TmbisxRqUCl6jt2uyJspR4+k38MeWQYsfPOtv6VM3N2uOOhewMaQ35qBK8kKn8EuzKP4ouNYTdXxvJgmCfrwRPrHZ621wkip910dISpAHjJQNHwL/tDEY+YMKcWMnO1CupjKBB8uIiED0pVE4W0eULfmSPEFMuCPedwuiwoQQ8twj3pqCUYJ4NisAvwoCkRUHPbSdkwcaCfvSbzCY4vE4v21RB3KT6MPYeWa7rC3sDOQLza0w==</SESSION_KEY>

</CKYC_INQ>

<Signature xmlns="http://www.w3.org/2000/09/xmldsig#"><SignedInfo><CanonicalizationMethod

Algorithm="http://www.w3.org/TR/2001/REC-xml-c14n-20010315"/><SignatureMethod

Algorithm="http://www.w3.org/2001/04/xmldsig-more#rsa-sha256"/><Reference

URI=""><Transforms><Transform Algorithm="http://www.w3.org/2000/09/xmldsig#envelopedsignature"/></Transforms><DigestMethod

Algorithm="http://www.w3.org/2001/04/xmlenc#sha256"/><DigestValue>xTMdnPXRctasXp0ULxT/QzOfp

90=</DigestValue></Reference></SignedInfo><SignatureValue>jqGYE7V5QcwWFzuLi3E5XZaB/LgYK5Km

RU0Hy7T0sszo4UcS6NeetH36Iez/PnW9N2tYyKyfvBxY4FuIywOzwdeFjXTizaItYsIfBKBPevBq7u8lc44DkrwT

G7gJ06UqkkXzeoxje54bdlPSbl9adXeH5AIM2geh3WB/snrus7ZyZCSAng9H6LrY7F3AJJCe55S/HpBxmgCran

93vYu2FpBJ4iK3rkxuJwwMWcwD7CG2tWxMr66UyOzTSirhs8995ceByzQHp2LiBnghFImytyR4PryIcG1lDJC

/tYgAe3ZdVZZ4TvX6tjKRWOZiGBwdHQXQcEFCotDxtVbVYqyKVA==</SignatureValue></Signature></REQ_ROOT>

Note: Separate Signature XML tag should be created like above once the request is signed.

10) Verify the signature XML object from the above request using FI's public key [This key is stored in CKYC server when user uploaded initially from upload public key page]

Convert the signature which is in string to document using

javax.xml.parsers.DocumentBuilderFactory and org.w3c.dom.Document libraries

Get the public keys details using java.security.cert.CertificateFactory and java.security.cert.Certificate libraries

Validate the signature which is in document with the public key using javax.xml.crypto.dsig.dom.DOMValidateContext library.

API error messages:

1. Public key is not available in the server. Please upload the public key on the CKYC application.

Error if public key is not available:

2. <<IP Address>>: IP is not registered on CKYCRR portal

Error if IP is not registered:

3. <<IP Address>>: IP does not match with IP registered on CKYCRR portal

Wrong IP address given in the request:

4. Please enter Institution Code

API request should contain FI code which is registered with CKYC application, error if failed:

5. Digital signature cannot be verified. The certificate is expired.

FI uploaded public key certificate expired.

6. Invalid expiry date in Public Key

FI uploaded public key certificate is corrupted and system is unable to read the expiry details.

7. Session Key Decryption error. Please encrypt with correct public key

Reason 1: FI is using incorrect cersai public key

Reason 2: Session key generated incorrect

Reason 3: Session key encrypted using CERSAI public key is incorrect.

8. PID Data Decoding error

Reason 1: PID data encrypted using session key is incorrect.

Reason 2: Session key generated incorrect

Reason 3: Framed PID is incorrect

9. Request XML not in proper format:

Incorrect framing of xml tags is being sent to CKYC API.

10. Digital signature cannot be verified. Please sign the document with proper key

Reason 1: If User signed the request xml with wrong pattern of signature tags.

Reason 2: FI Public key and Private key does not match.

Reason 3: Opening and closing tag of PID_DATA values are not proper.

Reason 4: If xml version tag is missing in PID_DATA [Example: - <?xml version="1.0" encoding="UTF-8"

standalone="yes"?>]

Reason 5: If PID_DATA contains extra spaces between child xml tags.

11. Your Request Could Not Be Processed. Please try after some time

Users can connect with helpdesk team by sharing the REQUEST details for checking with TECH Team.